

Poisson Models

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Intended Learning Outcomes

Students will be able to:

- Fit and interpret a generalized linear model of family Poisson
- Validate a Poisson model
- Explain when an alternative model should be fitted
- Fit and interpret a quasi-Poisson and negative binomial models

Fitting a Poisson GLM

- Number of eggs laid and female body size of vapourer moth:

1. Distribution of response = Poisson

2. Predictor function:

$$\text{Number of Eggs} = \beta_0 + \beta_1 X \text{ body length}$$

3. Link between the predictor and the mean of the distribution: log-linear:

$$\log(\text{Number of Eggs}) = \beta_0 + \beta_1 X \text{ body length}$$



Moth Eggs Example

Call:

```
glm(formula = Eggs ~ BodyLength, family = "poisson", data = motheegs)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-2.51406	0.17138	-14.67	<2e-16	***
BodyLength	1.17594	0.02724	43.17	<2e-16	***

$$\log(\text{Number of Eggs}) \\ = -2.51 + 1.18X \text{ body length}$$

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 3151.44 on 38 degrees of freedom
Residual deviance: 236.31 on 37 degrees of freedom
AIC: 421.19

Number of Fisher Scoring iterations: 5

Interpreting Coefficients

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-2.51406	0.17138	-14.67	<2e-16	***
BodyLength	1.17594	0.02724	43.17	<2e-16	***

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$$\log(\text{Number of Eggs}) = -2.51 + 1.18X \text{ body length}$$

$$\text{Number of Eggs} = e^{-2.51+1.18X \text{ body length}}$$

- For every cm increase in body length, the number of eggs increases by a natural log factor of 1.18 **OR**
- For every cm increase in body length, the number of eggs increases by a factor of $e^{1.18}$ or 3.25-fold.  **EFFECT SIZE**

Fold Change

- For fold changes if e^{β_1} is greater than 1 = positive effect, if less than 1 negative effect

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-2.51406	0.17138	-14.67	<2e-16	***
BodyLength	1.17594	0.02724	43.17	<2e-16	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

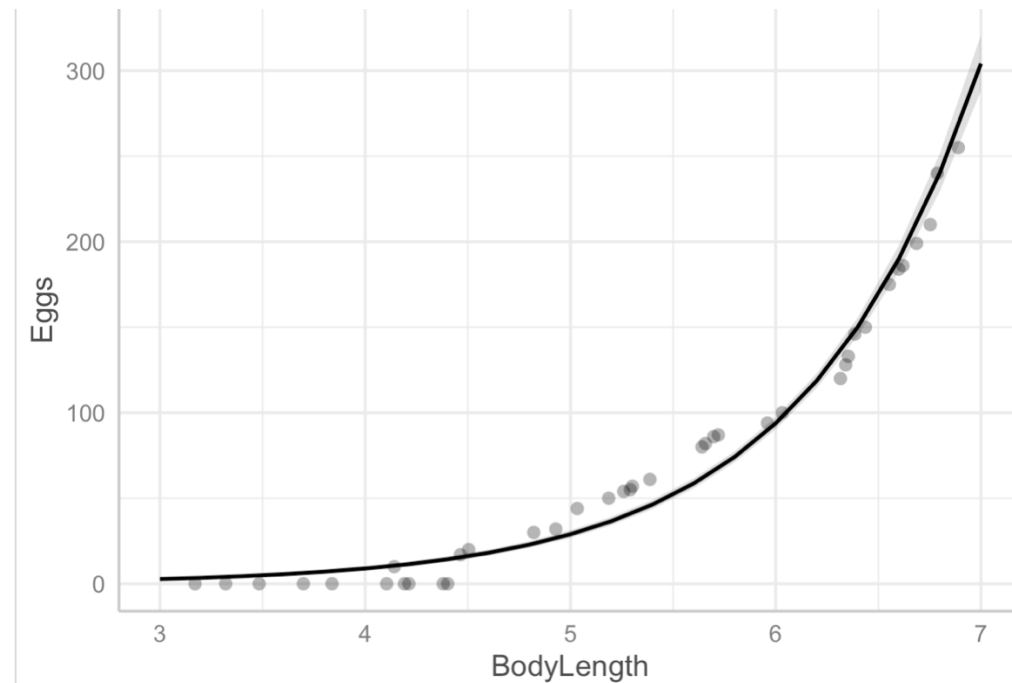
- $e^{1.18} = 3.25$ fold (positive effect)
- $e^{-0.13} = 0.31$ fold (negative effect)

Why Fold Change?

- For every cm increase in body length, the number of eggs increases by a factor of $e^{1.18}$ or 3.25-fold???
- The fitted slope is not the same across the body lengths – it is multiplicative.

$$\text{Number of Eggs} = e^{-2.51 + 1.18X \text{ body length}}$$


$$\text{Number of Eggs} = e^{-2.51} * e^{1.18X \text{ body length}}$$



Quick Maths

- At body length = 0cm

$$\text{Number of Eggs} = e^{-2.51} * e^{1.18 \times 0}$$

$$\text{Number of Eggs} = e^{-2.51}$$

$$\text{Number of Eggs} = 0.08$$

- At body length = 1cm

$$\text{Number of Eggs} = e^{-2.51} * e^{1.18 \times 1}$$

$$\text{Number of Eggs} = e^{-2.51} * e^{1.18 \times 1}$$

$$\text{Number of Eggs} = 0.08 * 3.25$$

$$\text{Number of Eggs} = 0.26$$

Quick Maths

- At body length = 0cm

$$\text{Number of Eggs} = e^{-2.51} * e^{1.18 \times 0}$$

$$\text{Number of Eggs} = e^{-2.51}$$

$$\text{Number of Eggs} = \mathbf{0.08}$$

- At body length = 1cm

$$\text{Number of Eggs} = e^{-2.51} * e^{1.18 \times 1}$$

$$\text{Number of Eggs} = e^{-2.51} * e^{1.18 \times 1}$$

$$\text{Number of Eggs} = \mathbf{0.08} * 3.25$$

$$\text{Number of Eggs} = 0.26$$

Moth Eggs Example

Call:

```
glm(formula = Eggs ~ BodyLength, family = "poisson", data = motheegs)
```

Coefficients:

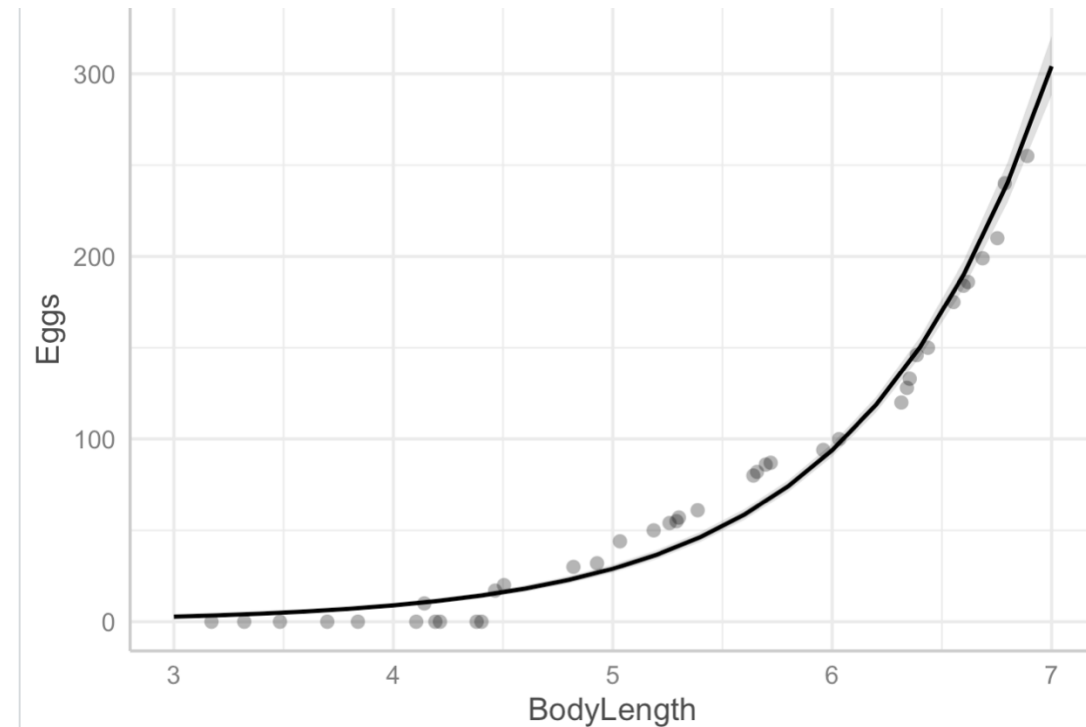
	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.51406	0.17138	-14.67	<2e-16 ***
BodyLength	1.17594	0.02724	43.17	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 3151.44 on 38 degrees of freedom
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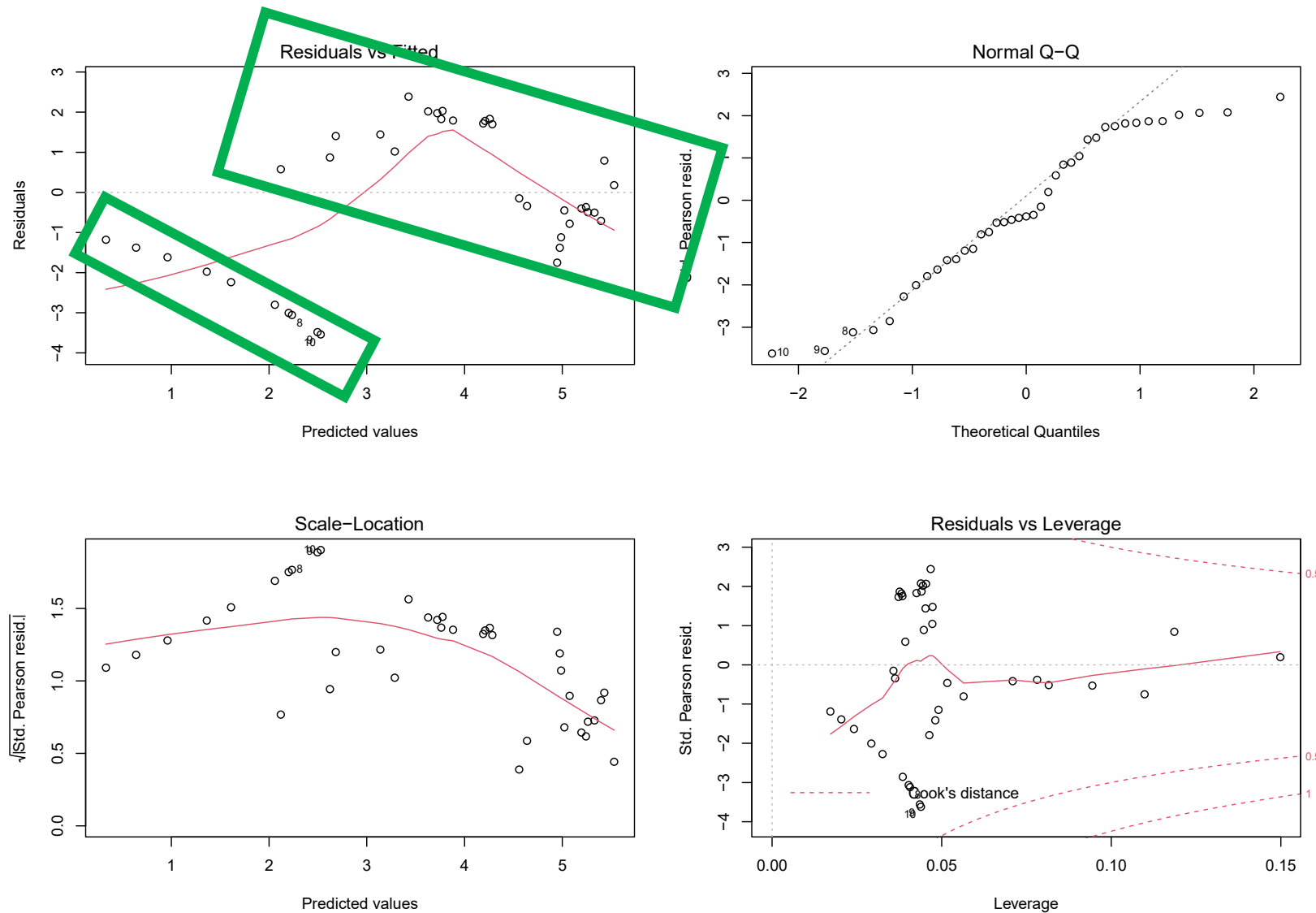
Null & Residual Deviance

Null deviance: 3151.44 on 38 degrees of freedom
Residual deviance: 236.31 on 37 degrees of freedom

- Null summarises the deviance/variance explained by a null model:
 - $\log(\text{Number of Eggs}) = 1$ – an intercept only model
 - $\text{lm}(\text{Eggs} \sim 1)$
- Residual deviance/variance not explained by your model → the error
- Both used to estimate goodness-of-fit for model
- Pseudo- R^2 : $1 - \left(\frac{\text{Residual Deviance}}{\text{Null Deviance}} \right)$
 - $1 - (236.31 / 3151.44) = 0.93$

93% of the variation in the number of eggs can be explained by female moth body length

Model Validation - Diagnostics



Dispersion

(Dispersion parameter for poisson family assumed to be 1)

- Dispersion refers to the theoretical or expected amount of variability based on Poisson distribution **assumption.**
- Overdispersion is very common in Poisson models and can be due to a whole host of factors:
 - Too simplistic (missing explanatory variables and/or interaction terms)
 - Explanatory variables measured on different scales
 - A covariate has a non-linear effect
 - One or more outliers
 - Zero inflation
 - Inherent dependency in the data – i.e. pseudoreplication

Dispersion Parameter

- Tells us how much larger or smaller is our conditional variance to our conditional mean
- Dispersion parameter should equal 1
 - >1 overdispersion
 - <1 underdispersion

Null deviance: 3151.44 on 38 degrees of freedom

Residual deviance: 236.31 on 37 degrees of freedom

$$236.31/37=6.89$$

- What level of dispersion is acceptable? **Still debated**
- So, what next?

Accounting for Overdispersion

- Quasi-likelihood methods
 - Assumes overdispersion only impacts the standard errors
 - Standard errors are adjusted by scaling

Call:
glm(formula = Eggs ~ BodyLength, family = "poisson", data = motheeggs)

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.51406	0.17138	-14.67	<2e-16 ***
BodyLength	1.17594	0.02724	43.17	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

Null deviance: 3151.44 on 38 degrees of freedom
Residual deviance: 236.31 on 37 degrees of freedom
AIC: 421.19

Number of Fisher Scoring iterations: 5

Call:
glm(formula = Eggs ~ BodyLength, family = "quasipoisson", data = motheeggs)

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-2.51406	0.35093	-7.164	1.72e-08 ***
BodyLength	1.17594	0.05578	21.082	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

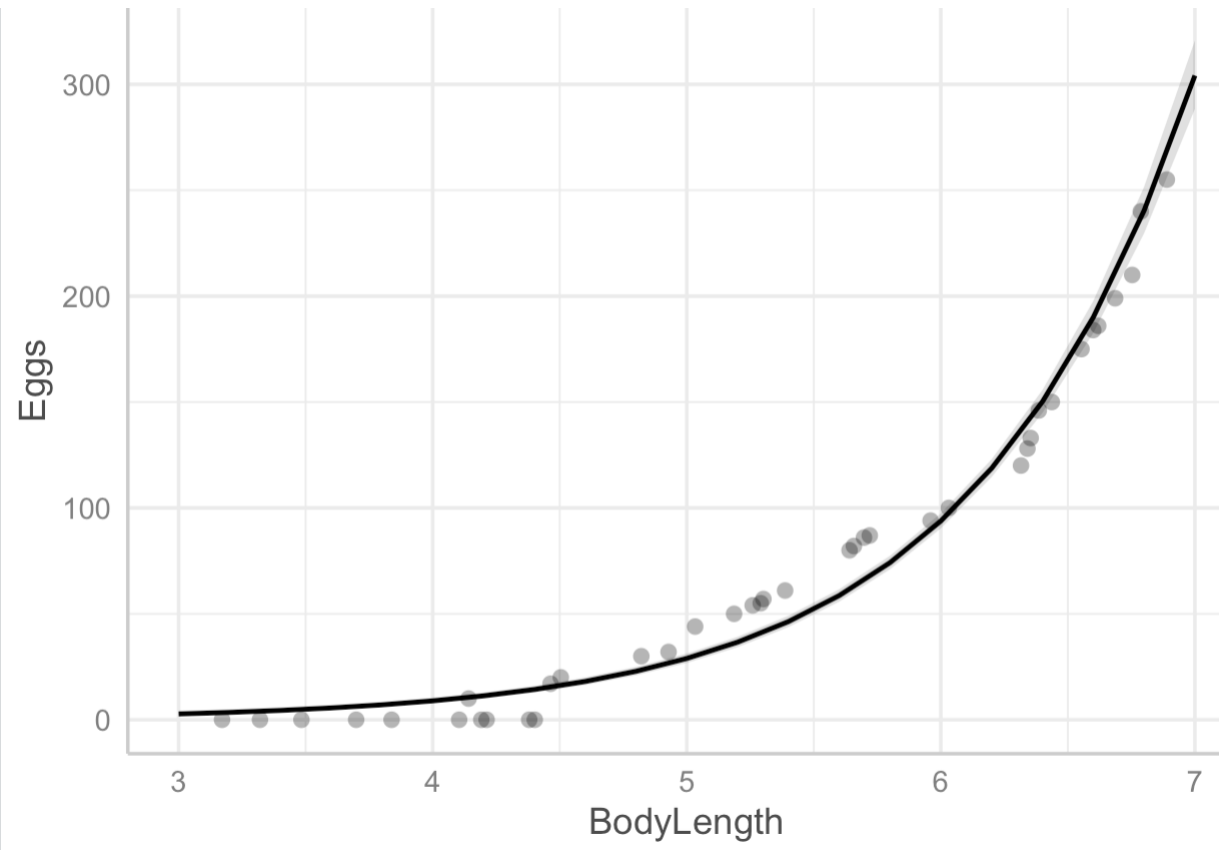
(Dispersion parameter for quasipoisson family taken to be 4.192907)

Null deviance: 3151.44 on 38 degrees of freedom
Residual deviance: 236.31 on 37 degrees of freedom
AIC: NA

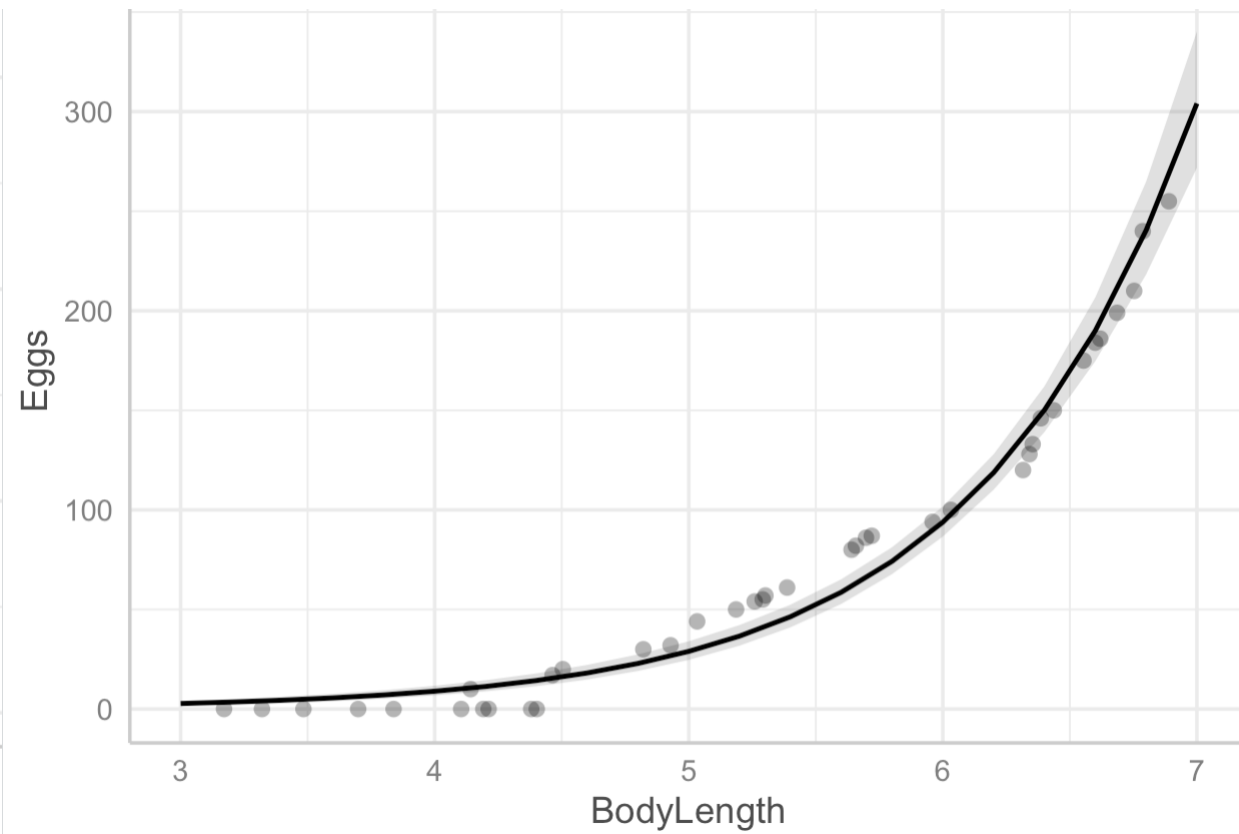
Number of Fisher Scoring iterations: 5

Accounting for Dispersion

Poisson



Quasi-Poisson



Accounting for Overdispersion – Negative Binomial Models

- Negative binomial approach
- Essentially, an overdispersed Poisson distribution converges with a negative binomial distribution.
- A negative binomial will give you different estimates and standard errors.
- Model Equation:

$$\text{Number of Eggs} = e^{-2.51+1.18X \text{ body length}}$$

- Pseudo- $R^2 = 1-(236.3/3151/1)$
= 0.93

Call:

```
glm.nb(formula = Eggs ~ BodyLength, data = motheeggs, init.theta = 647267.306,  
link = log)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.51414	0.17139	-14.67	<2e-16 ***
BodyLength	1.17595	0.02724	43.16	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for Negative Binomial(647265.2) family taken to be 1)

Null deviance: 3151.1 on 38 degrees of freedom
Residual deviance: 236.3 on 37 degrees of freedom
AIC: 423.19

Number of Fisher Scoring iterations: 1

Recap of overdispersion

- Might want to explore other options:
 - Too simplistic (missing explanatory variables and/or interaction terms)
 - Explanatory variables measured on different scales
 - A covariate has a non-linear effect
 - One or more outliers
 - Zero inflation
 - Inherent dependency in the data – i.e. pseudoreplication
- If you're interested look up:
 - Offsets
 - Zero inflated and truncated models
 - Generalised linear mixed models

1. "A Beginner's Guide to GLM and GLMM with R" – Zuur, Hilbe & N leno (2015 - Book)
2. "A brief introduction to mixed effects modelling and multi-model inference in ecology" Harrison *et al* (2018- Paper)

Summary

- Poisson models with the log-linear link function are able to handle Poisson data that are positive integers
- The estimates are on this log-linear scale and require exponentiating, which expresses the slope coefficient as a factor due to the multiplicative nature
- Poisson models can be validated by examining the diagnostic plots and the dispersion parameter
- Overdispersion can be caused by a whole host of factors
- A Quasi-likelihood approach or a negative binomial model can be fitted to account for overdispersion